

KaranEDU

AI-Powered Offline-First Learning Assistant

Artificial Intelligence Final Project | SDG 4: Quality Education

Student: Karan Lingayat	SDG Goal: 4 – Quality Education	Platform: Android 8.0+ v1.0.0	Year: 2026
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1. Selected SDG and Reason for Selection

Selected Goal: SDG 4 — Quality Education

SDG 4, adopted as part of the United Nations 2030 Agenda, calls for inclusive and equitable quality education and lifelong learning opportunities for all. It targets elimination of gender and wealth disparities, free primary and secondary schooling, and dramatically expanded access to affordable higher and vocational education.

Despite significant global progress over the past two decades, a severe learning crisis persists. Over **258 million children and youth** remain out of school, and a further **2.9 billion people** — nearly 37% of the world population — lack reliable internet access, excluding them from most modern digital learning tools. Premium tutoring apps, coaching services, and AI platforms remain unaffordable for the majority of students in low- and middle-income countries. This project directly addresses that gap with a completely free, offline-capable AI tutor that works on any Android device.

2. Problem Statement

What is the problem?

Millions of students — especially in rural areas, developing nations, and low-income households — cannot access quality, personalised academic support. Existing AI tutoring tools require stable internet connectivity and, in many cases, paid subscriptions, systematically excluding those who need help most.

Where does this problem exist?

The gap is most acute in Sub-Saharan Africa, South and Southeast Asia, and rural Latin America, but also affects underserved communities in developed nations. In India alone, fewer than 30% of rural households have reliable broadband, yet students there compete in the same high-stakes national examinations as urban peers.

Who is affected?

Secondary and higher-secondary students preparing for board and competitive examinations, first-generation learners without access to private coaching, and teachers in under-resourced schools lacking supplementary materials.

Why is it serious?

Without adequate academic support, students from disadvantaged backgrounds consistently under-perform relative to their potential. This perpetuates intergenerational poverty, widens income inequality, and wastes enormous human capital. The UN projects that if current trends continue, SDG 4 targets will not be met by 2030.

3. Proposed Solution

KaranEDU is a free, open-source Android application that delivers AI-powered personalised tutoring on any smartphone — with or without internet. It integrates **Google Gemini AI** for intelligent responses, **Room SQLite** for offline storage, and **Firebase Firestore** for cloud sync when connectivity returns.

How the system works:

- Student registers with Gmail via Firebase Auth and sets up their academic grade and focus area.
- Any question entered in the Chat Tutor is sent to the Gemini API via Kotlin Coroutines.
- The AI response is returned and simultaneously cached in Room SQLite for offline replay.
- Quiz questions, flashcard sets, and study plans are also stored locally.
- Firebase Firestore syncs all local data across devices when connectivity is restored.

KaranEDU is designed for students from Grade 8 through undergraduate level, particularly those preparing for board exams, entrance tests, or seeking concept clarity — completely free and without advertisements.

4. Project Features

AI Chat Tutor (Gemini Powered)

Conversational AI tutor answering any academic question with grade-appropriate explanations, mnemonics, step-by-step breakdowns, and resource suggestions. All responses are cached in Room DB for full offline access.

AI Quiz Generator

Automatically generates custom multiple-choice quizzes on any subject and topic. Provides instant per-question feedback, detailed explanations, and a diagnostic scorecard stored locally and synced to Firestore.

Spaced-Repetition Flashcards

AI-generated bite-sized flashcards presented as interactive 3D flip cards. Students rate each card (Easy / Medium / Hard) to drive an automatic spaced-repetition review schedule that maximises long-term retention.

Smart Daily Study Planner

Structured daily checklists for Mathematics, Physics, Essay Writing, and other subjects. Task completion persists locally across app restarts and offline sessions, with a real-time progress counter.

Offline-First SQLite Cache & Cloud Sync

Room SQLite stores all data on-device indefinitely. Firebase Firestore performs background synchronisation when the device reconnects, ensuring seamless cross-device continuity.

5. Technology Stack

Layer	Technology	Purpose
UI / Frontend	Kotlin + Jetpack Compose (M3)	Declarative adaptive Android UI
Architecture	MVVM + ViewModel	Lifecycle-aware state management
AI / NLP	Google Gemini SDK	Chat tutoring, quiz & flashcard generation
Async	Kotlin Coroutines + StateFlow	Non-blocking API calls & live updates

Local Storage	Room SQLite (Jetpack)	Offline-first persistent data cache
Authentication	Firebase Auth (OAuth / Gmail)	Secure sign-in & user identity
Cloud Sync	Firebase Cloud Firestore	Cross-device data synchronisation
Build	Gradle + Android Studio	Project build tooling & IDE

6. Application Screenshots

The screenshots below are taken directly from the working KaranEDU Android application, demonstrating each major feature in operation.

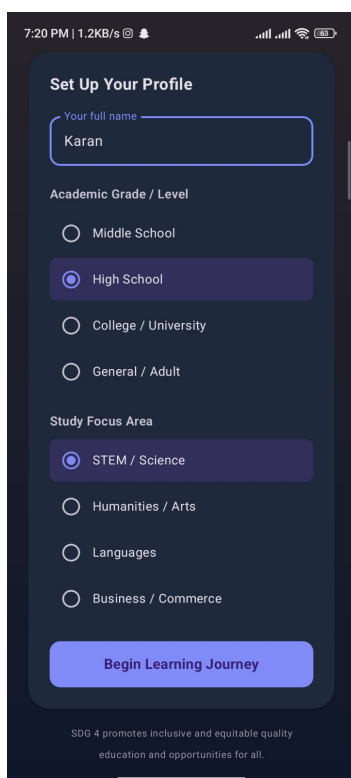


Fig 1: Login Screen – Cloud Sync & Gmail Auth

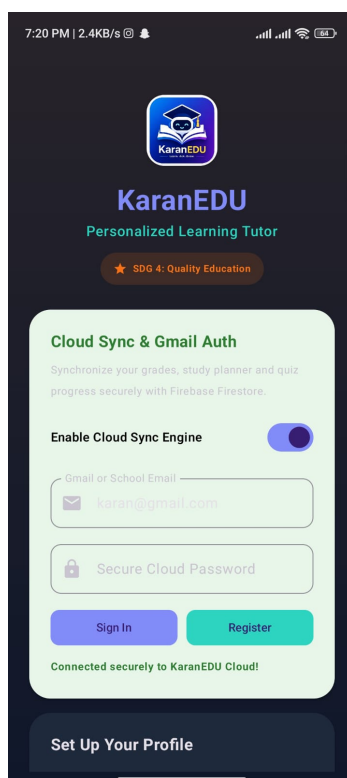


Fig 2: Profile Setup – Grade & Focus Area Selection

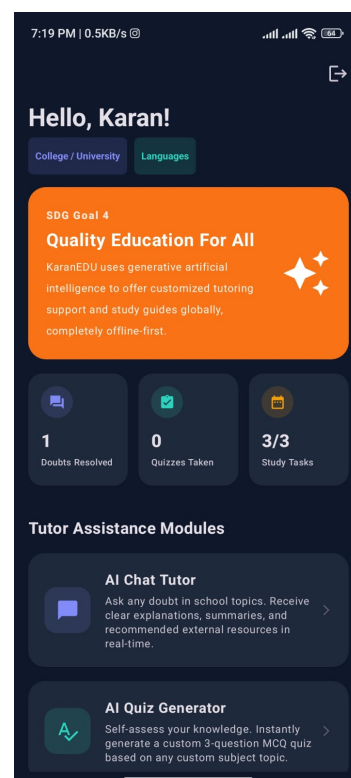


Fig 3: Main Dashboard – Tutor Assistance Modules

Left to right: The Login screen shows Gmail-based Cloud Sync authentication with Firebase. The Profile Setup screen allows students to choose their academic grade (Middle School through College) and study focus area (STEM, Humanities, Languages, Business). The Dashboard presents statistics — Doubts Resolved, Quizzes Taken, Study Tasks — and the four main AI modules.

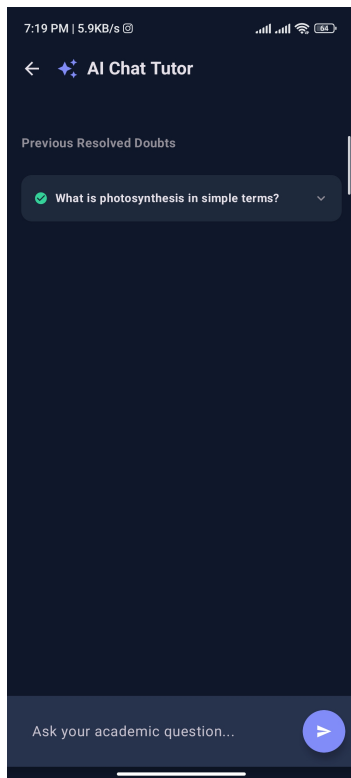


Fig 4: AI Chat Tutor – Previous Resolved Doubts

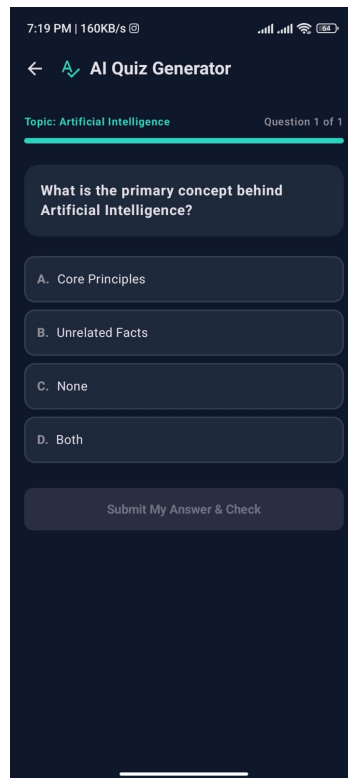


Fig 5: AI Quiz Generator – MCQ Screen

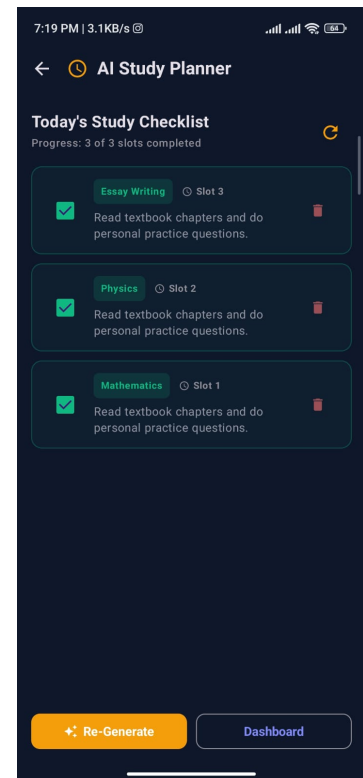


Fig 6: AI Study Planner – Daily Checklist

Left to right: The AI Chat Tutor shows a cached resolved doubt ("What is photosynthesis in simple terms?") with an input field for new questions. The AI Quiz Generator displays an auto-generated MCQ on Artificial Intelligence with four options and a progress bar. The AI Study Planner shows a fully completed daily checklist (3/3 slots) across Essay Writing, Physics, and Mathematics, with options to Re-Generate or return to the Dashboard.

7. Open Source Repository and Download Links

Resource	URL
GitHub Repository	https://github.com/karan942006/KaranEDU
APK Direct Download	https://github.com/karan942006/KaranEDU/raw/refs/heads/main/KaranEDU.apk
Project Website	https://karan942006.github.io/KaranEDU/

The GitHub repository contains the full Kotlin source under MVVM architecture, a README with build and setup instructions, the Room database schema, and guidance for Gemini API key configuration and Firebase integration. The project is open-source under the MIT licence. The APK can be sideloaded directly on any Android 8.0+ device without requiring the Play Store.

8. Future Scope

Multi-Language Support

Adding Hindi, Tamil, Telugu, Bengali, and other regional languages will open KaranEDU to a vastly wider demographic, addressing literacy barriers that English-only platforms cannot reach.

iOS Application

A native Swift / SwiftUI version will extend reach to Apple device users, particularly in markets where iPhones are the dominant personal device.

Voice-Based Interaction

Speech-to-text input and text-to-speech output will assist students with reading difficulties or visual impairments and enable hands-free studying during commutes.

Teacher Dashboard and Classroom Mode

A teacher-facing portal for assigning quizzes, tracking class-wide progress, and identifying struggling students using aggregate diagnostic data from Firestore.

Adaptive Learning Engine

A lightweight on-device ML model that analyses quiz and flashcard performance to dynamically adjust content difficulty, topic weighting, and study-plan recommendations for truly personalised learning paths.

National Curriculum Alignment

Mapping content to CBSE, ICSE, State Boards (India) and IB / Cambridge IGCSE frameworks for structured, exam-specific preparation across different boards.